

Report Assignment 1 - RM

Group 2

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Introduction

In this study, we analyze and hedge a portfolio of interest rate derivatives using the discount curve already obtained via bootstrap from liquid market instruments.

The portfolio consists of two positions. The first is a long payer swaption, with notional 650 Mln €, with a 10-year and 1-month expiry written on a forward starting interest rate swap with a tenor of 5 years. The second is a vanilla 10-year interest rate swap (IRS), with notional 550 Mln €, where the fixed leg is received. The floating leg is indexed to 3-month Eubor, which follows the ACT/360 convention, while the fixed leg follows a 30/360 convention. In both cases, the fixed leg is paid annually.

At first glance, the portfolio appears to be exposed to rising interest rates over the first 10-year horizon and to declining rates over the subsequent 5 years. This intuition stems from the combination of a receiver position in the 10-year IRS, which benefits from lower rates, and a long payer swaption on a forward-starting 5-year swap, which provides exposure to higher rates in the forward period.

This intuition will be proved rightful in our work.

1 Question 1: Mark-to-Market

The mark-to-market value of the portfolio can be defined as its current value, computed with today's market data.

It is computed as the sum of:

- The **swaption price**, obtained via the Black model;
- The **IRS value**, computed as the difference of the NPVs of floating and fixed legs.

To compute them, we first obtained the forward swap rate and the swaption price per unit of notional, as well as the IRS MtM per unit notional. Both prices were then multiplied by the corresponding notional to get the final value of the portfolio.

Forward Swap Rate	Swaption Price (per unit)	IRS MtM (per unit)	Portfolio MtM (€)
5.3002%	0.115955	-0.000000	75.370.456,41

Table 1: Summary of key valuation quantities

The sign and magnitude of the MtM are consistent with the current level of interest rates relative to the swaption strike and the IRS fixed rate. In particular, since the swaption is at-the-money, its value is driven primarily by volatility and time to maturity, while the IRS value reflects deviations from the par swap rate.

Moreover, the IRS has a mark-to-market close to zero, as it is valued at the par swap rate, which implies that the present value of the fixed leg is approximately equal to that of the floating leg.

As a consequence, the overall portfolio value is almost entirely driven by the swaption component.

2 Question 2: Numerical DV01

The DV01 (Dollar Value of a Basis Point) is a standard measure of interest rate risk, defined as the change in the portfolio value resulting from a 1 - basis point parallel shift in the yield curve. It provides a linear approximation of the impact of rate movements on the portfolio MtM.

In practice, the DV01 is computed through a full revaluation approach: market rates are shifted by 1 basis point, the discount curve is re-bootstrapped, and the portfolio is re-priced. The difference between the shocked and initial values represents the DV01.

Quantity	Value (€)
Portfolio DV01	-367.999,97

Table 2: Portfolio DV01 under a parallel shift

The negative sign of the DV01 is consistent with the composition of the portfolio. In particular, the portfolio is dominated by the receiver IRS, which behaves like a fixed coupon bond and therefore loses value when interest rates increase.

Although the payer swaption benefits from rising rates, its sensitivity is smaller compared to that of the IRS. As a result, the overall portfolio exhibits a net short position in interest rates, leading to a negative DV01.

3 Question 3: Analytical DV01 Approximation

An analytical approximation of the portfolio DV01 can be obtained by decomposing the portfolio into its main risk components.

The swaption contribution is approximated through its delta with respect to the forward swap rate. This delta measures the sensitivity of the swaption price to changes in the underlying forward rate, assuming that discount factors remain unchanged. As discussed during lectures, this hypothesis is not really accurate and realistic.

The IRS is instead approximated using a duration-based approach. In particular, a receiver IRS can be interpreted as the difference between a fixed coupon bond and a floating rate bond. The fixed leg consists of deterministic cash flows, discounted using the term structure, and therefore behaves exactly like a standard fixed coupon bond. The floating leg, being periodically reset to the current market rate as rates move, is not exposed to interest rates risk.

As a consequence, the overall sensitivity of a receiver IRS is mainly driven by the fixed leg, which justifies approximating its interest rate exposure using the Macaulay duration of a corresponding fixed coupon bond.

By combining these two components, an approximate DV01 of the portfolio is obtained as the sum of the swaption delta contribution and the IRS duration-based sensitivity, with negative sign.

However, this analytical approximation differs from the numerical DV01. The main reason is that the delta of the swaption captures only the sensitivity with respect to the forward swap rate, while ignoring the impact of interest rate changes on discount factors. In contrast, the numerical DV01 is computed via full revaluation after re-bootstrapping the entire yield curve, thus incorporating the effect of rate shifts on all maturities. This justifies the results obtained:

Measure	Value (€)
Portfolio DV01 (Proxy)	-294.118,50
Portfolio DV01 (Numeric)	-367.999,97
Difference	73.881,47

Table 3: Comparison between analytical and numerical DV01

4 Question 4: Delta Hedging with a 10y IRS

In this section, we construct a hedging strategy aimed at neutralizing the interest rate risk of the portfolio. The goal is to eliminate the portfolio sensitivity to a parallel shift of the yield curve, as measured by the DV01.

The hedge is implemented using a 10-year interest rate swap (IRS), which is chosen due to its liquidity and its exposure to the relevant part of the yield curve. We implement the hedging using both the numerical DV01 and the approximated one.

4.1 Hedging Strategy

The hedging approach is based on a first-order approximation of the portfolio value with respect to interest rates. For small changes in rates, the variation in the portfolio value can be approximated as:

$$\Delta V \approx DV01 \cdot \Delta r$$

Derivation of the Hedge Notional N_H

Let:

- V_P : Market value of the existing portfolio.
- V_H : Market value of the hedging instrument (e.g., a Receiver IRS).
- N_H : Notional amount of the hedging instrument.
- v_H : Unit value of the hedging instrument (value per €1 of notional).
- r : Interest rate.

The total value of the hedged position is defined as:

$$V_{total} = V_P + V_H = V_P + N_H \cdot v_H \quad (1)$$

To achieve delta neutrality, the sensitivity of the total value with respect to the interest rate must be zero:

$$\frac{dV_{total}}{dr} = 0 \implies \frac{dV_P}{dr} + N_H \frac{dv_H}{dr} = 0 \quad (2)$$

Solving for the hedge notional N_H :

$$N_H = -\frac{\frac{dV_P}{dr}}{\frac{dv_H}{dr}} \quad (3)$$

By definition, the **DV01** is the change in value for a 1 basis point (1bp = 0.0001) move in rates, hence:

$$DV01_{ptf} = \frac{dV_P}{dr} \cdot 1bp \quad (4)$$

$$DV01_{irs} = \frac{dv_H}{dr} \cdot 1bp \quad (5)$$

Substituting these definitions into the expression for N_H , the 1bp terms cancel out, leading to the standard hedging formula:

$$N_H = -\frac{DV01_{ptf}}{DV01_{irs}} \quad (6)$$

By constructing a hedged portfolio with zero DV01, the first-order sensitivity to interest rate changes is eliminated. As a result, small parallel shifts in the yield curve produce negligible changes in the portfolio value.

For the analytical approximation, we followed the same procedure using the approximated DV01 values, such that:

$$DV01_{ptf} = \Delta_{swaption} \cdot N_{swaption} - DU_{irs} \cdot N_{irs} \quad (7)$$

$$DV01_{irs} = -DU_{irs} \cdot 1bp \quad (8)$$

4.2 Results

The hedge notional is then obtained using the formula above and subsequently rounded to the nearest €1 million, reflecting market trading conventions.

The required notional is:

Hedging Approach	Required IRS Notional (€)	New Overall Notional (€)	Residual DV01 (€)
Numerical hedge	-459.000.000	91.000.000	31,00
Analytical hedge	-356.000.000	194.000.000	—

Table 4: Comparison between numerical and analytical hedging strategies

4.3 Interpretation of the Hedge Sign

The sign of the hedge notional provides important information about the direction of the portfolio exposure to interest rates.

In our case, the portfolio DV01 is negative, meaning that the portfolio loses value when interest rates increase. To offset this exposure, a hedging instrument with positive DV01 is required.

A payer IRS has a positive DV01, as its value increases when interest rates increase. Therefore, to hedge a negative DV01, it is necessary to take a payer position (i.e. reduce the notional of the receiver IRS).

More generally, the sign of the hedge follows directly from the condition:

$$DV01_{ptf} + N_H \cdot DV01_H = 0$$

which implies that the hedge must have opposite sensitivity to that of the original portfolio.

This ensures that the combined portfolio is locally insensitive to small parallel shifts in interest rates.

Moreover, one can notice how the analytical approximation significantly **underestimates** the required hedge notional.

5 Question 5: Coarse-Grained Bucket DV01

The coarse-grained bucket DV01 approach provides a more detailed representation of interest rate risk compared to a standard parallel DV01, as it captures sensitivities across different segments of the yield curve and allows for more effective hedging of non-parallel movements for different scenarios.

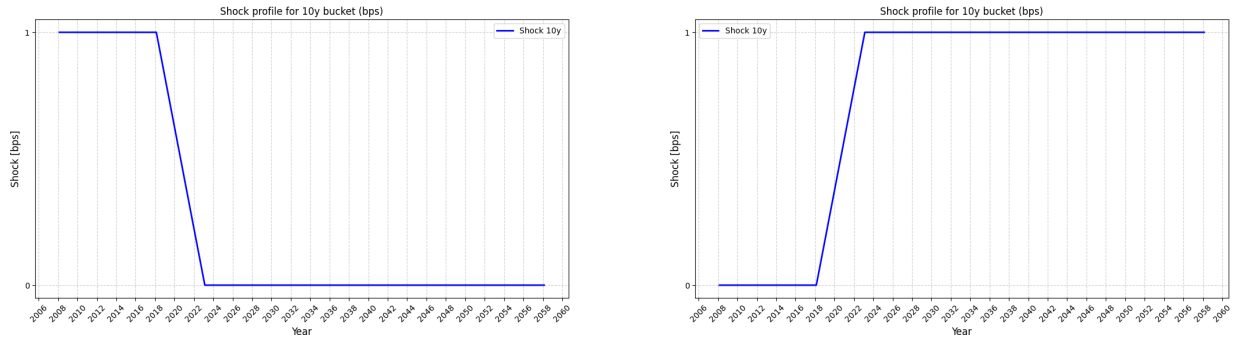


Figure 1: Applied shock for the two Coarse-Grained Buckets

The methodology is based on applying non-uniform shocks to the market instruments used in the bootstrap. We applied the shocks for the 10y and 15y macro buckets as in the figures above.

Formally, for each bucket j , the portfolio sensitivity is defined as:

$$DV01^{(j)} = V_{\text{shocked}}^{(j)} - V$$

where V is the initial portfolio value and $V_{\text{shocked}}^{(j)}$ is the portfolio value obtained after applying the bucket-specific shock and re-bootstrapping the yield curve.

The results obtained are:

Bucket	DV01 (€)
10y Bucket	-925.075,12
15y Bucket	557.584,75

Table 5: Coarse-grained bucket DV01 of the portfolio

The results highlight a markedly different sensitivity of the portfolio across the two buckets. In particular, the portfolio exhibits a strong negative exposure in the 10-year bucket and a positive exposure in the 15-year bucket.

This pattern can be explained by the composition of the portfolio and it's consistent with out initial intuition. In the 10y bucket, both instruments contribute to the sensitivity, but the IRS exposure is dominant, generating a strong negative contribution.

In contrast, the 15y bucket exposure is driven solely by the payer swaption, as the 10-year IRS has no sensitivity beyond its maturity. The underlying swap of the swaption spans the 10y–15y interval, and therefore a rate increase in this segment raises the value of the swaption. This explains the positive DV01 observed in the 15-year bucket.

Overall, the opposite signs of the bucket DV01s reveal a significant exposure to changes in the slope of the yield curve, confirming that a simple parallel DV01 hedge is not sufficient to fully neutralize the portfolio risk.

6 Question 6: Hedging with Two IRS

In this section, we improve the hedging strategy by moving from a single-instrument parallel DV01 hedge to a two-instrument bucket hedge. The objective is to neutralize the portfolio sensitivity in the two coarse-grained buckets identified in the previous section, namely the 10-year and 15-year buckets.

To do so, we select two liquid IRS among those used in the bootstrap, with maturities close to the relevant regions of the curve. In our case, the natural choice is to use a 10-year IRS and a 15-year IRS, since they provide direct exposure to the two macro-buckets where the portfolio sensitivity is concentrated.

6.1 Methodology

For each hedging IRS, we compute the coarse-grained bucket DV01s by re-pricing the instrument under the same 10y- and 15y-specific shocks scenarios used for the portfolio. Denoting by $DV01_{ptf}^{(10)}$ and $DV01_{ptf}^{(15)}$ the portfolio bucket sensitivities, and by $DV01_{IRS_1}^{(10)}$, $DV01_{IRS_1}^{(15)}$, $DV01_{IRS_2}^{(10)}$, and $DV01_{IRS_2}^{(15)}$ the sensitivities of the two hedging swaps, the hedge notionals N_1 and N_2 are obtained by solving the linear system

$$\begin{cases} DV01_{ptf}^{(10)} + N_1 DV01_{IRS_1}^{(10)} + N_2 DV01_{IRS_2}^{(10)} = 0 \\ DV01_{ptf}^{(15)} + N_1 DV01_{IRS_1}^{(15)} + N_2 DV01_{IRS_2}^{(15)} = 0 \end{cases}$$

This formula follows directly from the hedging condition requiring the total bucket sensitivity of the hedged portfolio to be zero in both macro-buckets.

6.2 Results

Solving this system provides the notionals that simultaneously offset both exposures:

Position	Instrument	Notional (€)
New position	IRS 10y	-1,154.000.000
New position	IRS 15y	517.000.000
Total position	IRS 10y	-604.000.000
Total position	IRS 15y	517.000.000

Table 6: IRS positions after two-instrument hedging strategy

The hedge requires reversing the original 10-year IRS exposure and taking a large opposite position, while introducing a new position in the 15-year IRS to offset the long-end sensitivity of the portfolio.

This makes the two-instrument hedge more accurate than the single-IRS hedge, especially in the present case, where the portfolio exhibits opposite sensitivities in the 10-year and 15-year buckets. Finally, as in the previous question, the hedge notionals are rounded to the nearest €1 million to reflect market trading conventions.

To be sure that our choice was correct, we also computed the hedge for some different swaps, as shown in the following results, just to discover that the initial choice is the less onerous hedging.

Strategy	IRS 1 Notional	IRS 2 Notional
IRS 12y / IRS 15y	-1.674.000.000	1,088.000.000
IRS 1y / IRS 15y	-9.178.000.000	517.000.000
IRS 4y / IRS 12y	-4.843.000.000	1.513.000.000

Table 7: Alternative Hedging strategies

7 Question 7: P&L under Curve Flattening

In this section, we assess the effectiveness of the hedging strategies by evaluating the portfolio P&L under a non-parallel movement of the yield curve. In particular, we consider a flattening scenario characterized by a +1 basis point shift in the 10-year bucket and a -1 basis point shift in the 15-year bucket.

The scenario is implemented by constructing a combined shock profile that increases rates in the 10-year region while decreasing them in the 15-year region. The yield curve is then re-bootstrapped under this shock, and the portfolio is re-priced accordingly.

Formally, the P&L of a given portfolio configuration is defined as:

$$\text{P\&L} = V_{\text{shocked}} - V_{\text{initial}}$$

where V_{initial} is the value of the hedged portfolio under the original curve and V_{shocked} is the value obtained after applying the flattening shock and re-bootstrapping the curve.

We evaluate the P&L for two different hedging strategies:

- a hedge based on a single 10-year IRS, constructed to neutralize the parallel DV01;
- a hedge based on two IRS (10y and 15y), constructed to neutralize the coarse-grained bucket DV01s.

From a theoretical perspective, the single-instrument hedge is based on eliminating only the first-order sensitivity to parallel shifts. Therefore, it is not expected to perform well under non-parallel movements, such as flattening or steepening of the yield curve.

In contrast, the two-instrument hedge is designed to neutralize the portfolio sensitivity in each bucket separately. By eliminating both the 10y and 15y bucket DV01s, the hedged portfolio becomes locally insensitive to movements affecting the slope of the curve between these maturities.

As a result, the two-instrument hedge provides a more robust protection against the considered flattening scenario. This highlights the importance of using bucket-based hedging techniques when the portfolio exhibits significant exposure to different regions of the yield curve. This is easily confirmed by our results:

Hedging Strategy	Initial Value (€)	Value After Shock (€)	P&L (€)
10y IRS Hedge	75.370.456,41	74.256.407,81	-1.114.048,60
10y + 15y IRS Hedge	75.370.456,41	75.812.412,24	441.955,84

Table 8: P&L under flattening scenario for different hedging strategies

The results clearly show that the hedge based on a single 10-year IRS performs poorly under the flattening scenario, generating a significant loss. This is consistent with the fact that such a hedge neutralizes only the parallel DV01 and does not account for changes in the slope of the yield curve.

In contrast, the hedge constructed using two IRS, 10y and 15y, provides a much more effective protection, even resulting in a positive P&L. This confirms that bucket-based hedging strategies are better suited to manage non-parallel shifts, as they explicitly target the sensitivity of the portfolio across different segments of the curve. The Coarse-Grained hedging strategy proves to be superior in non-parallel shift scenarios, such as the tested curve flattening.

To further enhance accuracy, a Full Bucket-DV01 (eg. 1y, 2y, 5y, 10y, 15y, 20y...) strategy could be implemented. However, such a granular approach involves a significant trade-off between hedging precision and increased operational and transaction costs, primarily due to the complexity of managing a larger number of swap positions.

Code Errors

- **Swaption Payer/Receiver Logic:** In the `swaption_price_calculator` function, the valuation formulas for Payer and Receiver swaptions were inverted.
- **Swap MtM Multipliers:** Also here, in the `swap_mtm` function, the valuation formulas for Payer and Receiver swaptions were inverted.
- **Swaption Maturity:** The swaption maturity months parameter was initially set to 0. This was corrected to 1.
- **IRS Type:** The IRS type was incorrectly defined as a "Payer" swap. This was corrected to "Receiver".
- **Date Handling:** Instances of the `today` variable were systematically replaced with `settlement_date` to account for the 2-days delay before a contract is fully operating.
- **Q2 and Q5** During the computation of the shocked portfolio Mark-to-Market, the base discount factors and base forward swap rates were mistakenly applied. These references were updated to explicitly use the shocked curves (`discount_factors_up` and `fwd_swap_rate_up`, along with their respective 10y/15y equivalents in Q5).
- **Q3** We changed the positive sign of the duration term into a negative one to correctly reflect the directional risk.
- **Q4** The terminology was updated to *underestimates* because the analytical approximation suggested a smaller payer swap notional was required to offset the initial position.